



SagarDrishti

A browser-native 3D digital twin of India's ocean

- **Problem Statement ID** - SIH26067
- **Problem Statement Title** - Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.
- **Theme** - Disaster Management
- **PS Category** - Software
- **Team ID** -
- **Team Name (Registered on portal)** - PixelPaws



Not a proposal on paper. 567 automated tests · 51 fps at 1080p on integrated graphics · 30,012 QC-passed instrument levels · zero requests leave the machine.

The Problem & Our Solution

SagarDrishti Bay of Bengal: model fields and the instruments in the water, on one screen, with a number for the gap between them.

THE PROBLEM · INCOIS'S OWN WORDS

- “No integrated, web-based 3D visualization platform currently exists” so forecasters toggle between disparate software packages, which impedes hazard assessment, search and rescue, fishery advisories and climate monitoring.

PROPOSED SOLUTION · HOW IT ADDRESSES THE PROBLEM

Volumetric column in a browser. 24 levels, 5 m to 2000 m.

All four instrument classes. 149 casts: temperature, salinity, oxygen, chlorophyll.

A skill number beside the field. Bias and RMSE per depth band.

Hazards drawn over the water. Live CAP v1.2 from NDMA SACHET.

A new source is config, not code. YAML plus a plugin. OGC WMS/WCS served.

WHERE THIS SITS

Nobody occupies the full cell.

DESKTOP

VOLUMETRIC

pyParaOcean · IDV

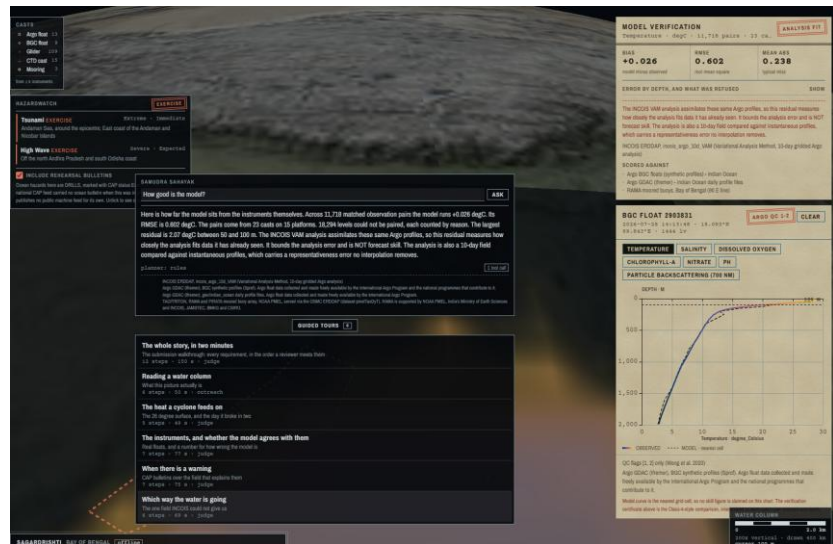
2D ONLY

ODV · Ferret / LAS

IN A BROWSER

SagarDrishti

MyOcean · Argovis



MODEL VERIFICATION		
Temperature · degC · 11,718 pairs · 23 ca...		
BIAS +0.026	RMSE 0.602	MEAN ABS 0.238
model minus observed	root mean square	typical miss
ERROR BY DEPTH, AND WHAT WAS REFUSED		

11,718 matched pairs

13,622 levels refused

0.602 degC RMSE

NetCDF · ASCII · CAP XML → source registry (YAML) → chunked Zarr cube → FastAPI + OGC WMS / WCS → WebGL 3D scene in the browser

1 SOURCES

INCOIS ERDDAP · LAS
Argo GDAC · Argo BGC
glider · CTD · RAMA
Copernicus GLORYS
NDMA SACHET (CAP)

2 INGEST

CF normalisation
QC flags 1 and 2
plugins: HF-radar, ADCP
ready
rechunk to Zarr
10.5 M values

3 DATA PLANE

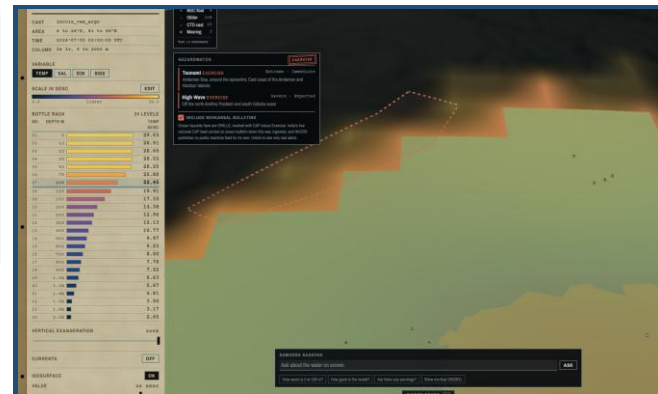
FastAPI · 13 routes
/field /profiles
/isosurface /currents
/scorecard /warnings
OGC WMS · WCS ·
interoperable

4 BROWSER

CesiumJS · WebGL2
depth slices, 24 levels
isosurface · current
vectors
colorbar · variable
time-step animation

SAMUDRA SAHAYAK · detachable service

calls the same public API. Cut this line and every P0 still
passes.



Working prototype: the 26 degC isosurface, marching cubes, offline.

5 VERIFICATION · Class-4 method

Interpolated to each cast's own position and depth. A level is refused, never filled.

7 ANY OGC CLIENT

Published as WMS 1.3.0 and WCS 1.0.0: a scalable, interoperable path into the tools
INCOIS already runs.

TECHNOLOGIES USED



Nextjs



Typescript



CesiumJS



ECharts



Python 3.11



FastAPI



xarray · Zarr



NumPy · SciPy

Docker

pytest

scikit-image

gsw (TEOS-10)

Playwright

Parquet

Standards: CF Conventions · OGC WMS 1.3.0 · OGC WCS 1.0.0 · OASIS CAP v1.2 · TEOS-10 · Argo QC (Wong 2020) · Class 4 (Ryan 2015)

FEASIBILITY AND VIABILITY



FEASIBILITY IS NOT A FORECAST HERE. THE SYSTEM ALREADY RUNS. EVERY FIGURE BELOW WAS MEASURED, NOT ESTIMATED

60-SEC DEMO

567

automated tests, 11 in a browser

51 fps

at 1080p on integrated graphics

0

requests leave the machine

10.5 M

gridded values, two agencies

30,012

QC-passed levels, 19 instruments

13

ocean parameters, 2 by plugins

ANALYSIS OF FEASIBILITY

- **Proven stack.** CesiumJS, FastAPI, Zarr.
- **Frame budget cleared.** 51 fps on the weakest GPU.
- **Reads INCOIS's own ERDDAP.** No duplication.
- **Zero licence cost.** One GPU server, open source.



THE SAME BUILD, ON A PHONE

- 1 variable selector
- 2 colorbar, min / max, log
- 3 24 levels, 5 m to 2000 m
- 4 vertical exaggeration
- 5 layer opacity
- 6 time-step animation

POTENTIAL CHALLENGES AND RISKS

DATA

- **Instrument coverage is uneven.** Not on INCOIS ERDDAP.
- **Datasets disagree on axes.** GLORYS stops at 1942 m.

TECHNICAL

- **Volumetric rendering is costly.** Translucent overdraw dominates.
- **Scaling to the full basin.** The national grid is larger.

ADOPTION AND TRUST

- **A wrong number beats no number.** Its caveat must travel.
- **A false alarm is unacceptable.** Drills are dashed.

STRATEGIES TO OVERCOME THEM

DATA

- **Parsers already written.** A file, not new code.
- **Never stretch one grid onto another.** The served depth is printed.

TECHNICAL

- **Depth-space level of detail.** At most 10 of 24 slices at once.
- **Chunked Zarr over range requests.** Cost is view-dependent.

ADOPTION AND TRUST

- **The caveat travels with the number.** On screen and in the API.
- **Four independent drill guards.** Exercise is refused.

Every control PS requirement F4 names. No install.

VIABILITY · WHY THIS SURVIVES AFTER THE HACKATHON

Native deployment path

INCOIS's own request. No data migrates.

Standards, not a silo

WMS 1.3.0 and WCS 1.0.0: their tools read it.

Maintainable by the owner

Standard Python and JavaScript.

Costs nothing to keep

Zero licences, one GPU server.

IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

DISASTER MANAGERS

A tsunami or high-wave CAP bulletin becomes a shape on the water.

OPERATIONAL FORECASTERS

"Can I trust today's field?" answered per depth band.

SEARCH AND RESCUE

Depth-resolved currents, which is what a drifting object follows.

FISHERY ADVISORIES

Subsurface temperature behind advisories reaching ~9 lakh households.

RESEARCH, EDUCATION, OUTREACH

Guided tours narrate the ocean for schools and exhibitions.

BENEFITS · SOCIAL, ECONOMIC, ENVIRONMENTAL

Rs 0

licence cost, against proprietary 3D suites

9 lakh

fisherfolk households reached by PFZ advisories

0.602

degC RMSE, with the analysis-fit caveat attached

NATIONAL ALIGNMENT

Digital India

Deep Ocean Mission

Blue Economy

SAGAR / MoES mandate

UN SDG 14: Life Below Water

Deployable on INCOIS infrastructure

The PS is an internal INCOIS tooling request, so the deployment story is native rather than argued.

WHAT CHANGES ON THE DAY IT MATTERS · A DEPRESSION FORMS IN THE BAY

1 SEE THE HEAT

Switch to the depth of the 26 degC isotherm, the layer a cyclone feeds on.

2 CHECK THE MODEL

The card says the field is 2.07 degC off across 50 to 100 m.

3 CONFIRM AGAINST THE WATER

Click the nearest Argo float. Its measured profile against the model.

4 PUT THE WARNING IN CONTEXT

The CAP bulletin is drawn over the water it concerns. Network off.

DETAILS AND LINKS OF THE REFERENCE AND RESEARCH WORK

METHOD AND VERIFICATION

- [Ryan et al. 2015: GODAE OceanView Class 4 verification, J. Op. Oceanogr. 8](#) · *the method behind our scorecard*
- [Wong et al. 2020: Argo data and its QC flags, Front. Mar. Sci. 7:700](#) · *why we keep only flags 1 and 2*
- [IOC, SCOR, IAPSO: TEOS-10 thermodynamic equation of seawater](#) · *our density field*
- [Balakrishnan Nair et al. 2013: INCOIS Ocean State Forecast validation, Current Science 105\(2\)](#) · *the Indian validation precedent*

WEB 3D OCEAN VISUALISATION

- [Qin et al. 2021: Web 3D framework for time-varying ocean forecast data, Env. Modelling & Software 135](#) · *the closest ancestor*
- [Yu, Qin & Xu 2025: WebGPU volume rendering of ocean scalars, Applied Sciences 15\(5\):2782](#) · *our rendering optimisations*
- [Tucker et al. 2020: Argovis, J. Atmos. Ocean. Tech. 37\(3\):401](#) · *the API contract we follow*
- [Blower et al. 2013: ncWMS, Env. Modelling & Software 47](#) · *OGC WMS over CF-NetCDF*
- [Signell & Pothina 2019: Cloud coastal model analysis, JMSE 7\(4\):110](#) · *why Zarr rather than whole NetCDF files*
- [pyParaOcean \(IISc\): desktop ocean volumetrics we bring to the browser](#)

DATA SOURCES WE INGEST

- [INCOIS ERDDAP: the analysis field, 24 levels, 5 to 2000 m](#)
- [Argo GDAC \(Ifremer\): Indian Ocean profile files](#)
- [Copernicus Marine: GLOBAL ANALYSISFORECAST_PHY_001_024 currents](#)
- [NOAA PMEL: RAMA moored buoy array, Indian Ocean](#)
- [NDMA SACHET: India's national CAP alert backbone](#)
- [INCOIS: Indian National Centre for Ocean Information Services](#)

OPEN STANDARDS IMPLEMENTED

- [CF Conventions for climate and forecast metadata](#)
- [OGC Web Service \(WMS\) 1.3.0](#)
- [OGC Web Coverage Service \(WCS\) 1.0.0](#)
- [OASIS Common Alerting Protocol \(CAP\) v1.2](#)

HOW THE RESEARCH CHANGED THE BUILD · not a bibliography, a set of decisions

Ryan 2015 → verify in observation space

Wong 2020 → QC flags 1 and 2 only

Yu 2025 → drop a slice that cannot change the picture

Signell 2019 → Zarr chunks over HTTP range reads

WHERE WE SIT · GAP ANALYSIS

Capability	Desktop tools	Web viewers	Digital Ocean	Sagar Drishti
Browser-native, no install	✗	✓	✓	✓
True volumetric water column	✓	✗	~	✓
Model + in-situ in ONE 3D scene	✗	✗	~	✓
Model-vs-observation skill on screen	✗	✗	✗	✓
CAP hazard layer in ocean context	✗	✗	✗	✓
Runs fully offline / air-gapped	✓	✗	✗	✓

Desktop tools: ODV, IDV, pyParaOcean. Web viewers: Argovis, MyOcean. INCOIS's Digital Ocean portal advertises 3D and 4D; what it does not do is fuse instruments into the volume or publish model skill beside the field.

OUR OWN WORK, AND HOW TO CHECK IT

567 automated tests, 11 in a real browser

6 dp TEOS-10 check values reproduced exactly

100% of panels cite dataset and timestamp

15 + 8 papers and existing tools reviewed for the gap analysis